CurricuPath Pilot Evaluation Report

Pilot evaluation (developer-side labels)

Project 26P73: AI-assisted Wits EIE learning pathway and module advisor

Evaluation protocol v2.5, pilot runs | 29 September 2026 | code version 5854b3df

Status: pilot, not a validated result. No independent academic reviewer has approved reference labels yet. Every score below is measured against developer-side answer keys, so it describes how the systems behave and differ. It is not evidence of independently validated accuracy. Figures published for other systems appear as context only.

Summary

- **What was run.** Five systems answered the same 60 synthetic EFA03/EFA04 eligibility cases once each, and the deterministic intent extractor was tested on 100 draft prompts.
- **How this differs from the frozen protocol.** One run per case instead of three; the model arms used gemini-3.5-flash-lite instead of the pinned gemini-3.6-flash; and the CurricuPath arms ran without Gemini intent extraction, so they show the deterministic path only.
- **CurricuPath against the handbook-RAG baseline.** Agreement with the developer labels (macro-F1) is 0.89 against 0.71 under the developer labels (difference +0.18, 95% interval +0.03 to +0.32; CurricuPath higher). Under the Senate Rules briefing labels it is 0.65 against 0.51 (inconclusive).
- **The answer key matters.** The two developer-side label sets agree on 78% of cases (kappa 0.67) and differ on 13, all on whether a corequisite that is only *planned* counts as satisfied. CurricuPath's macro-F1 moves from 0.89 to 0.65 between them.
- **CurricuPath makes the same decisions as the rules-only system on 50 of 60 cases.** On the other 10, rules-only is given the declared completeness fact as structured input and decides *ineligible*; CurricuPath does not reach a decision. Elsewhere any advantage over rules-only must come from language handling and clarification, which this case set does not test.
- **Intent understanding:** 85/100 correct (85%). The weak categories are career-path prompts and ambiguous module names.

1. Purpose and scope

The investigation asks whether hard eligibility gates and conversational clarification improve the correctness of Wits EIE curriculum advice compared with simpler strategies. The study is limited to eligibility decisions for the EFA03 (Electrical Engineering) and EFA04 (Information Engineering) programmes, each returned as one of three outcomes: *eligible*, *ineligible* or *manual review required*.

Under the frozen protocol (section 16), work without a named independent reviewer may continue only as a pilot and must not claim independently validated accuracy. This report is that pilot.

Its purpose is threefold: to demonstrate that the evaluation method works end to end on real system outputs, to identify failure patterns, and to isolate the one rule question (Rule 1.9, planned corequisites) that an academic reviewer has to settle. Claims about accuracy are deferred to the reviewer-labelled evaluation.

2. Method

2.1 Systems compared

System	What it does	Pilot configuration
CurricuPath	Intent extraction, module-name resolution, deterministic eligibility gates, and one scripted clarification where permitted.	Deterministic extraction; Gemini not used.
CurricuPath (no clarification)	The same system with clarification switched off.	Deterministic extraction; Gemini not used.
Rules only	The deterministic gates applied directly to the target module. No language layer.	No model.
Handbook RAG	Retrieves the 20 most similar handbook passages and asks one model to decide, with no rules supplied. The baseline representing existing handbook chatbots.	gemini-3.5-flash-lite, top-k 20.
Structured-rules LLM	As Handbook RAG, but the structured rules and source states are added to the prompt.	gemini-3.5-flash-lite, top-k 20.

2.2 Cases

60 synthetic profiles (no real student data), all drafted by the developers: 20 eligible, 20 ineligible and 20 manual review. Scenario types: academic discretion or unresolved record (20), corequisite unknown (10), prerequisite missing (10), prerequisite satisfied (20). Ten cases permit one scripted clarification.

2.3 Answer keys

Two developer-side keys were used. Neither is independent evidence.

- **Developer labels:** the expected outcome written when each case was drafted.
- **Senate Rules briefing labels:** a second reading of the same cases from a briefing document built from the 2026 Senate Rules. The document calls every label a draft and records no reviewer, so it is used only as a sensitivity check.

2.4 Measures

Three-class macro-F1 and balanced accuracy, with 95% intervals from a case-level bootstrap (10000 resamples, seed 20260918), and paired differences that share each resample. A difference is called directional only when its interval excludes zero; otherwise it is reported as inconclusive. A run with no valid decision counts as an error and stays in every denominator. Error types are reported separately: an unsafe false positive is a case that was not eligible but was answered eligible.

2.5 How the pilot departs from the frozen protocol

- One run per case instead of three, so repeatability is not measured.
- LLM arms used gemini-3.5-flash-lite, not the pinned gemini-3.6-flash, because of quota.
- CurricuPath arms ran without Gemini intent extraction, so they show the deterministic path only.
- An earlier Handbook-RAG attempt lost 27 of 60 runs to a per-minute quota limit; it was re-run paced and all 60 are valid.

3. Eligibility results

3.1 Agreement with the answer keys

System	Macro-F1, developer labels (95% CI)	Balanced accuracy	Macro-F1, briefing labels (95% CI)	Balanced accuracy	No valid decision
CurricuPath	0.89 (0.81 to 0.95)	0.83	0.65 (0.51 to 0.76)	0.70	10
Handbook RAG	0.71 (0.60 to 0.81)	0.72	0.51 (0.39 to 0.62)	0.54	0
Structured-rules LLM	0.80 (0.69 to 0.89)	0.80	0.66 (0.52 to 0.78)	0.77	0
Rules only	1.00 (1.00 to 1.00)	1.00	0.76 (0.65 to 0.86)	0.87	0
CurricuPath (no clarification)	0.89 (0.81 to 0.95)	0.83	0.65 (0.51 to 0.76)	0.70	10

The 10 cases with no valid decision. Rules only and both CurricuPath variants return no valid decision on the same 10 cases. These are the *corequisite unknown* cases, and they are also the 10 cases where one scripted clarification was permitted. The profile does not declare the corequisite, and the engine declines to decide rather than assume the module is not taken. The developers labelled all ten *ineligible*.

These are counted as errors by design, and they are not mapped to *manual review required*. The frozen protocol keeps *needs clarification* separate from manual review: clarification asks for missing information, whereas manual review means the system cannot decide even once the facts are supplied. It states that clarification is "never silently converted into manual review" and that failing to decide after the scripted reply is a decision error. Two caveats follow. The developer label of *ineligible* for these cases is itself open to a reviewer's challenge, and because the clarification step resolved none of them (section 6), the ten errors say more about the scripted reply than about the eligibility rules.

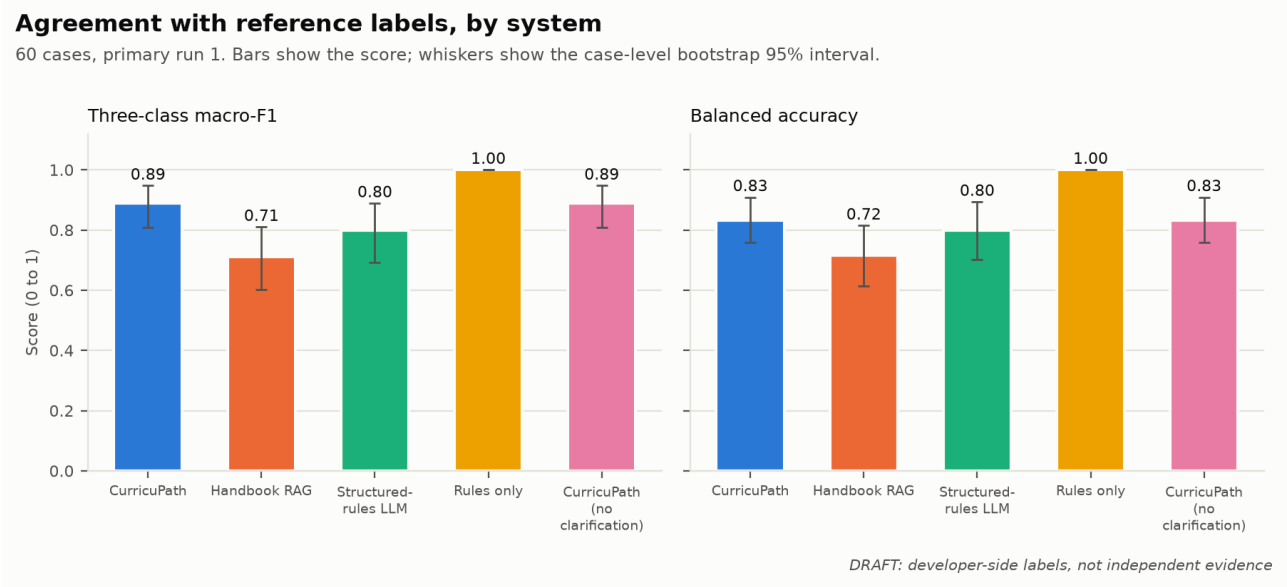


Figure 1. Agreement with the developer labels. Whiskers show the bootstrap 95% interval.

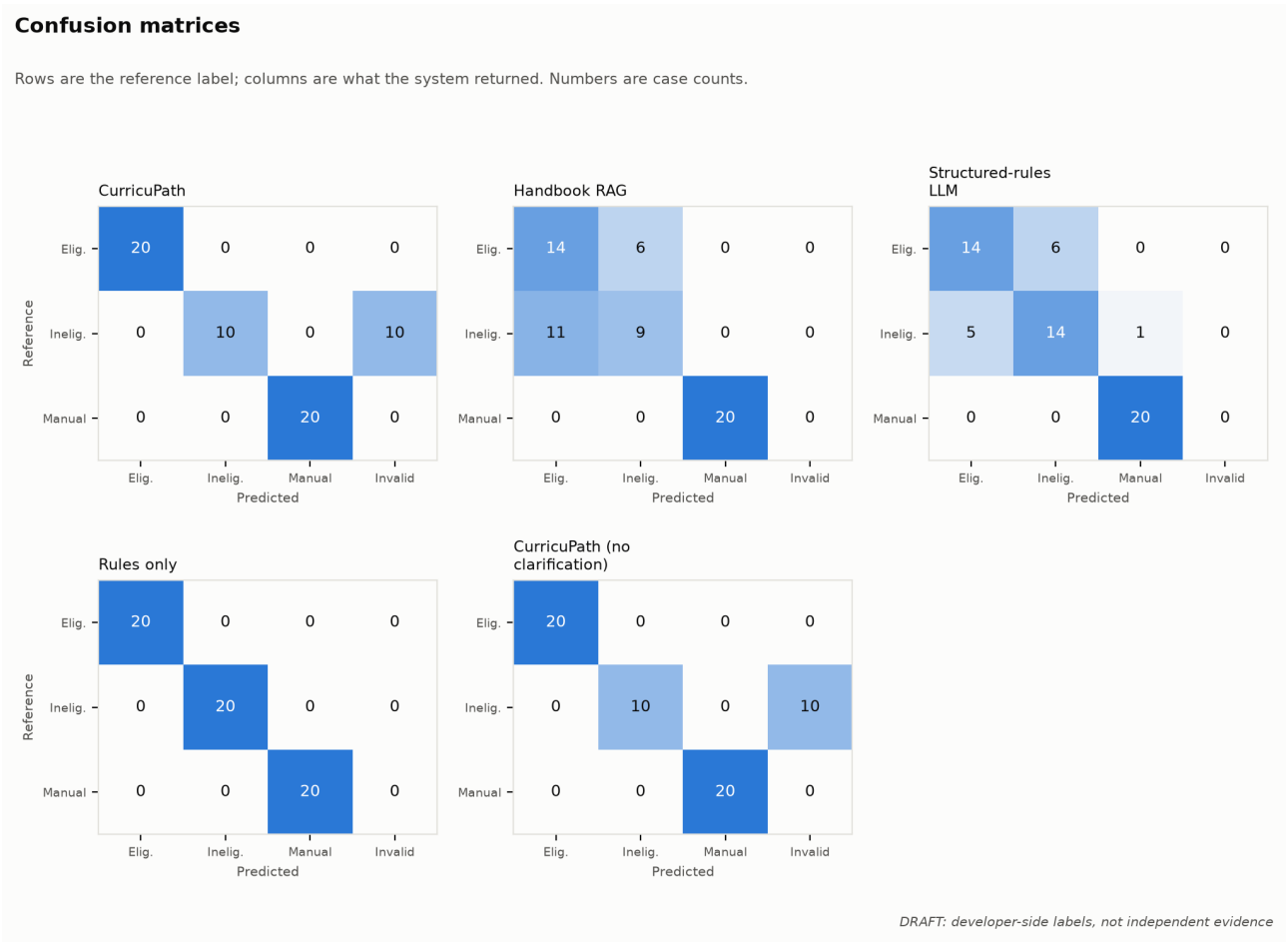


Figure 2. Confusion matrices against the developer labels. Rows are the reference label.

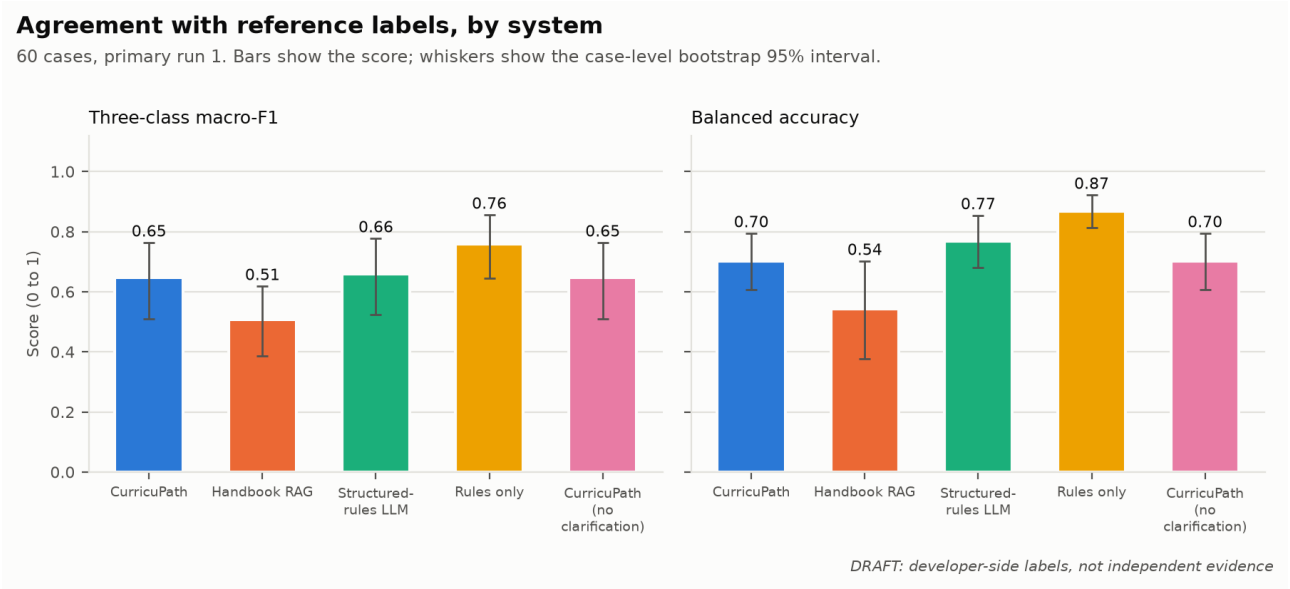


Figure 3. Agreement with the Senate Rules briefing labels.

3.2 Paired differences

Macro-F1 difference between CurricuPath and each other system on the same cases.

CurricuPath minus	Developer labels: difference (95% CI)	Reading	Briefing labels: difference (95% CI)	Reading
Handbook RAG	+0.18 (+0.03 to +0.32)	higher	+0.14 (-0.01 to +0.28)	inconclusive
Structured-rules LLM	+0.09 (-0.04 to +0.22)	inconclusive	-0.01 (-0.13 to +0.09)	inconclusive
Rules only	-0.11 (-0.19 to -0.05)	lower	-0.11 (-0.19 to -0.05)	lower
CurricuPath (no clarification)	+0.00 (+0.00 to +0.00)	identical	+0.00 (+0.00 to +0.00)	identical

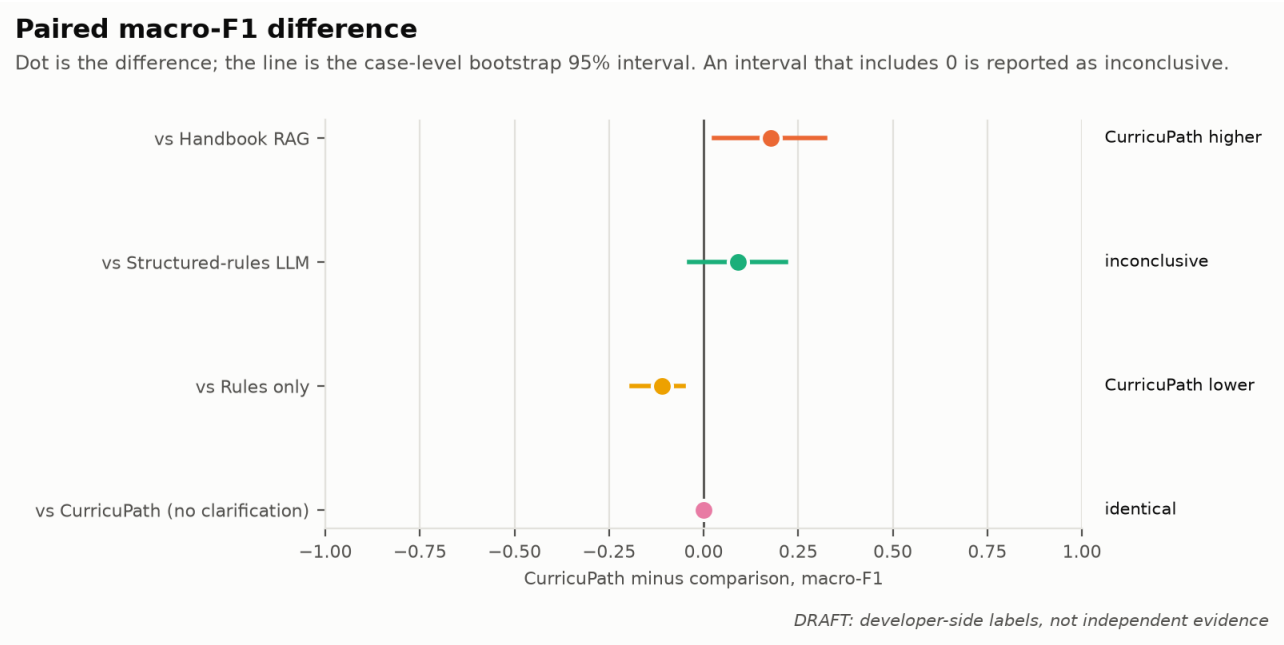


Figure 4. Paired macro-F1 differences against the developer labels.

3.3 Error types

Against the developer labels:

System	Unsafe false positive	Overly conservative	Correct escalation	Unsafe overconfidence	Unnecessary escalation
CurricuPath	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)
Handbook RAG	28% (11/40)	30% (6/20)	100% (20/20)	0% (0/20)	0% (0/40)
Structured-rules LLM	12% (5/40)	30% (6/20)	100% (20/20)	0% (0/20)	2% (1/40)
Rules only	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)
CurricuPath (no clarification)	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)

Against the Senate Rules briefing labels:

System	Unsafe false positive	Overly conservative	Correct escalation	Unsafe overconfidence	Unnecessary escalation
CurricuPath	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)
Handbook RAG	40% (21/53)	43% (3/7)	61% (20/33)	39% (13/33)	0% (0/27)
Structured-rules LLM	23% (12/53)	0% (0/7)	61% (20/33)	39% (13/33)	4% (1/27)
Rules only	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)
CurricuPath (no clarification)	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)

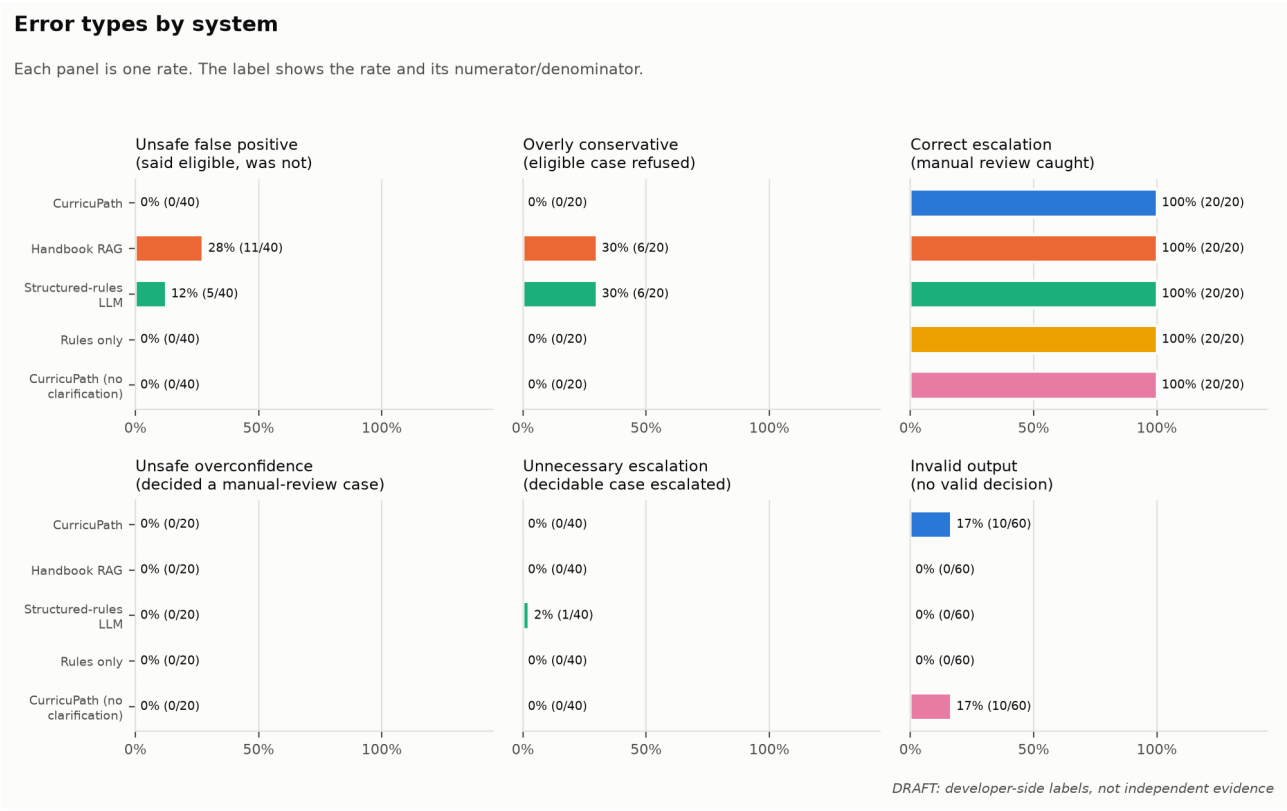


Figure 5. Error types against the developer labels.

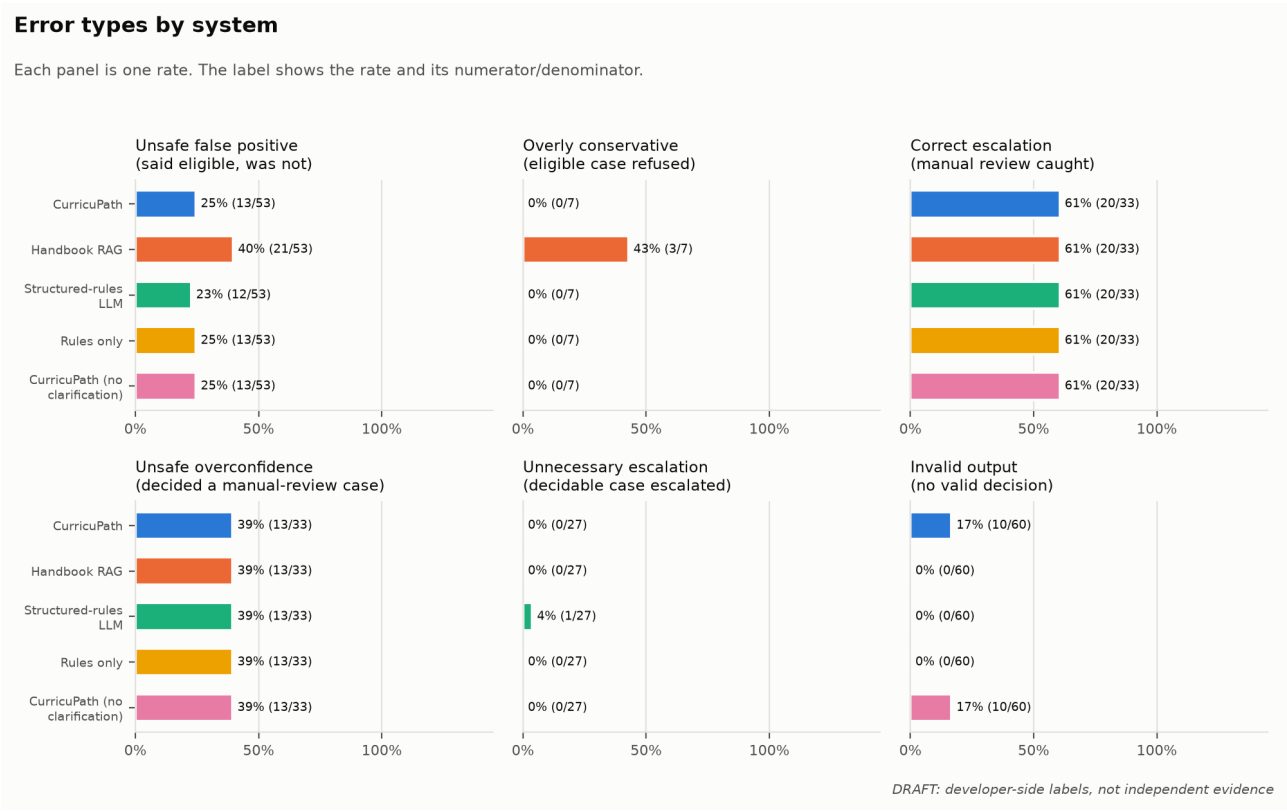


Figure 6. Error types against the Senate Rules briefing labels.

3.4 Sensitivity to the answer key

The two label sets agree on 78% of the 60 cases (Cohen's kappa 0.67). All 13 differences are the same: the developers marked the case *eligible* and the briefing marked it *manual review required*, because a corequisite appears only under the student's planned modules. Senate Rule 1.9 defines a corequisite as a course that must be taken with another, so whether a planned module counts is a question for an academic authority. The engine

currently accepts planned modules as satisfying a corequisite.

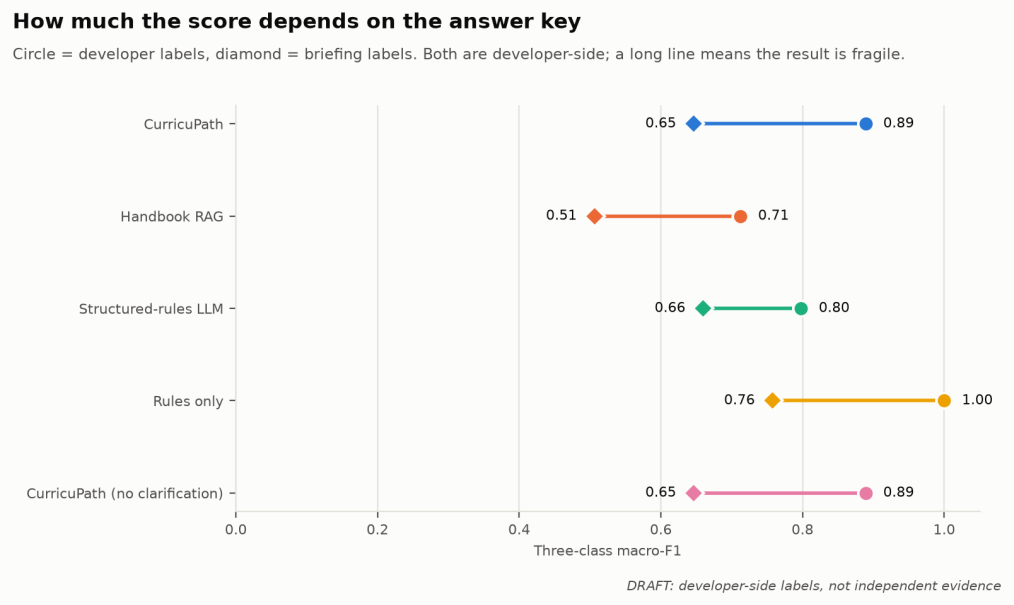


Figure 7. Macro-F1 under each answer key (circle: developer, diamond: briefing).

Cases where the label sets differ: ELIG001, ELIG002, ELIG003, ELIG005, ELIG006, ELIG007, ELIG010, ELIG013, ELIG014, ELIG015, ELIG018, ELIG019, ELIG020.

3.5 Response time

Times are per case, measured end to end through each system. The CurricuPath figures are for the deterministic path only; the Gemini-assisted path was not measured.

System	Median	95th percentile	Runs
CurricuPath	0.005 s	0.013 s	60
Handbook RAG	1.606 s	1.932 s	60
Structured-rules LLM	1.823 s	2.204 s	60
Rules only	0.001 s	0.001 s	60
CurricuPath (no clarification)	0.005 s	0.006 s	60

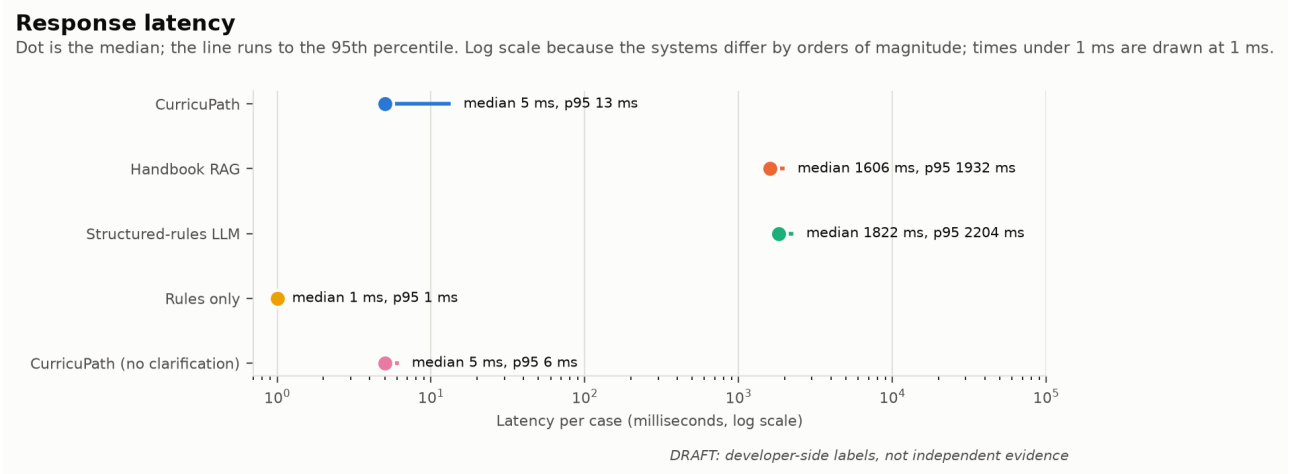


Figure 8. Response time by system (log scale).

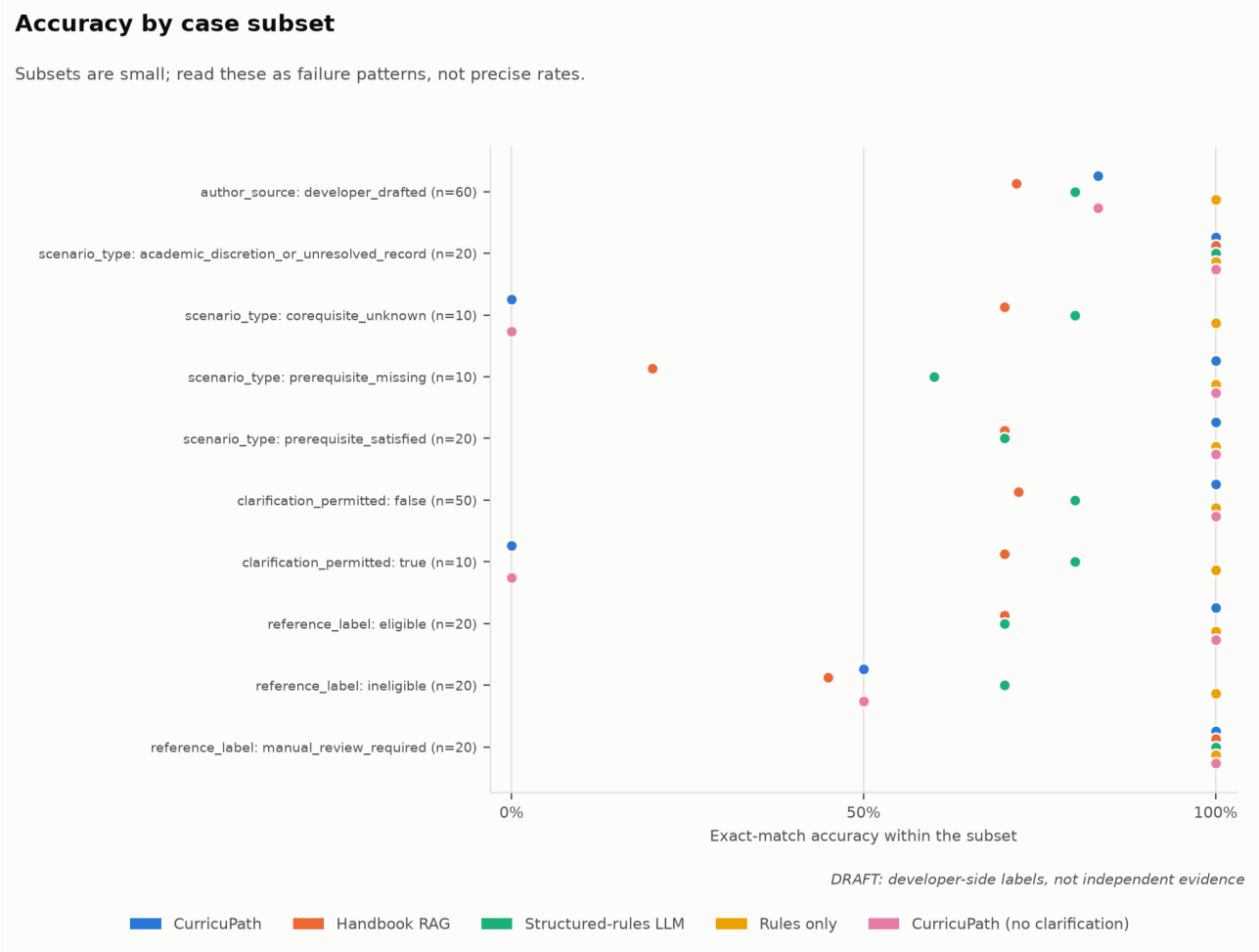


Figure 9. Exact-match accuracy within case subsets. Subsets hold 10 to 60 cases, so read these as failure patterns rather than precise rates.

4. Intent understanding

The deterministic extractor was run on 100 developer-drafted prompts that have not been independently reviewed. The Gemini-assisted extractor was not measured.

Prompt category	Correct	Rate	95% interval
ambiguous module name	0/10	0%	0.00 to 0.28
career path	6/10	60%	0.31 to 0.83
unknown or out of scope	4/5	80%	0.38 to 0.96
clarification follow up	5/5	100%	0.57 to 1.00
conversation control	5/5	100%	0.57 to 1.00
curriculum list	15/15	100%	0.80 to 1.00
eligibility	20/20	100%	0.84 to 1.00
module information	15/15	100%	0.80 to 1.00
public course search	15/15	100%	0.80 to 1.00
All prompts	**85/100**	**85%**	0.77 to 0.91

Where an entity was expected it was always extracted correctly: module code 35/35, programme code 18/18, year of study 12/12.

The ten *ambiguous module name* prompts are bare topic words such as electronics or signals, and this is the category where the extractor scores 0/10. The extractor itself was not changed. Before the fixes in section 4.2, a later layer did not recover these either: most were refused as out of scope. After the fixes, nine of the ten get a

question asking which module they mean; the tenth, circuits, matches only Electric Circuits and is answered directly (section 4.1). The other clear weakness is career-path prompts, four of ten of which were treated as course searches.

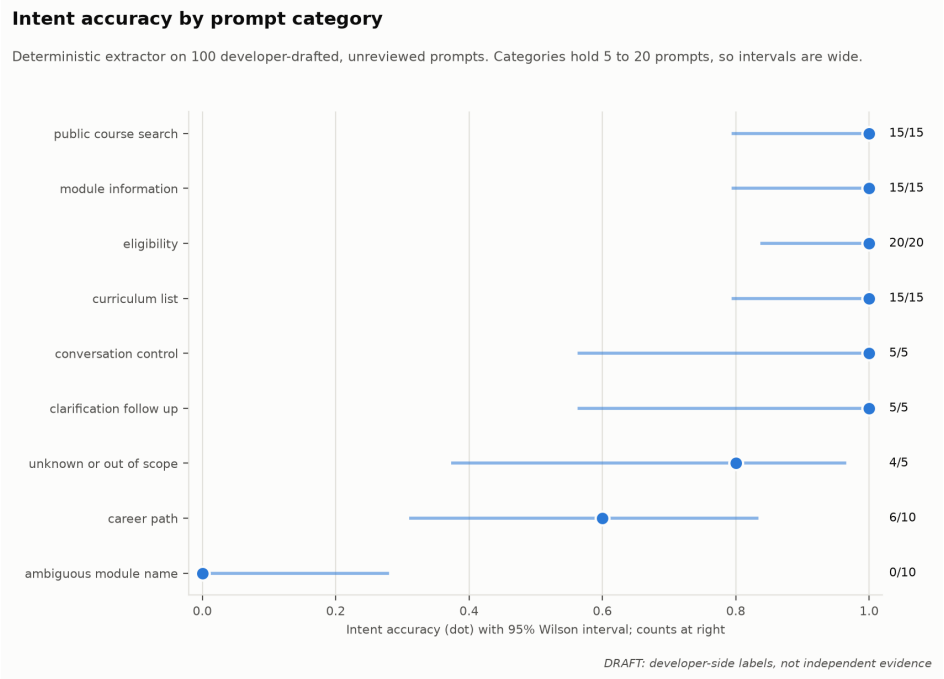


Figure 10. Intent accuracy by prompt category with 95% Wilson intervals.

4.1 Ambiguity and clarification (Experiment C, draft)

The 13 ambiguity cases were run with and without clarification. The fields that only the independent reviewer may supply (whether clarification is required, the scripted reply, and the accepted outcomes) are still pending, so only behavioural measures can be reported here.

Measure	Full CurricuPath	Without clarification
Initial intent matches the intended interpretation	13/13 (100%)	13/13 (100%)
Asked for clarification on the first turn	11/13 (85%)	0/13 (0%)
Answered on the first turn	2/13 (15%)	12/13 (92%)
Refused as out of scope on the first turn	0/13 (0%)	0/13 (0%)
Ended with an unresolved clarification	11/13 (85%)	0/13 (0%)

Clarification exists only in CurricuPath. In the version without it, the engine's clarification switch is off: it never asks, takes the best-scoring module when a name is shared, and otherwise gives a final "could not determine" answer. The other systems (rules only, Handbook RAG, structured-rules LLM) have no clarification step at all. Full CurricuPath asked in 11 of 13 cases; the version without clarification asked in 0. Whether those questions lead to better answers cannot be scored until the reviewer's scripted replies and accepted outcomes exist. These figures are after the fixes in section 4.2; before them, the correct intent was recognised in 3/13 cases, a question was asked in 2/13 and 8/13 were refused.

4.2 Why clarification failed, and the fixes

A diagnostic sweep asked about real modules at five levels of ambiguity (exact code, full name, name without its I/II marker, a single topic word, no topic) in four phrasings. Each prompt ran once with clarification switched off and once with it on, where any question received one simulated reply naming the module. These prompts were written by the developers to exercise the mechanism, not taken from the frozen cases. Clarification itself worked whenever it was reached; the failures were upstream:

- The scope gate refused module names as out of scope before module lookup could run.

- Question words such as "who teaches" were not stripped, so lookup found no candidates and the student got a generic question instead of a choice of modules.
- A name shared by several modules was only treated as a module question when it matched exactly, so the cases that most needed "which one?" never got it.

Three fixes addressed these, each with a regression test: question words are stripped before lookup; a short name shared by several modules becomes a module question that offers the candidates; and this decision runs before the scope gate, after prohibited requests are refused. A fourth planned fix, keeping state after the scope gate's own question, proved unnecessary. The 60 eligibility decisions were unchanged.

Named by	Studied set before	Studied set after	Unseen set before	Unseen set after
Full name	62%	100%	17%	100%
Name without I/II	22%	100%	17%	54%
Topic word	25%	100%	4%	50%
No topic	62%	100%	17%	100%

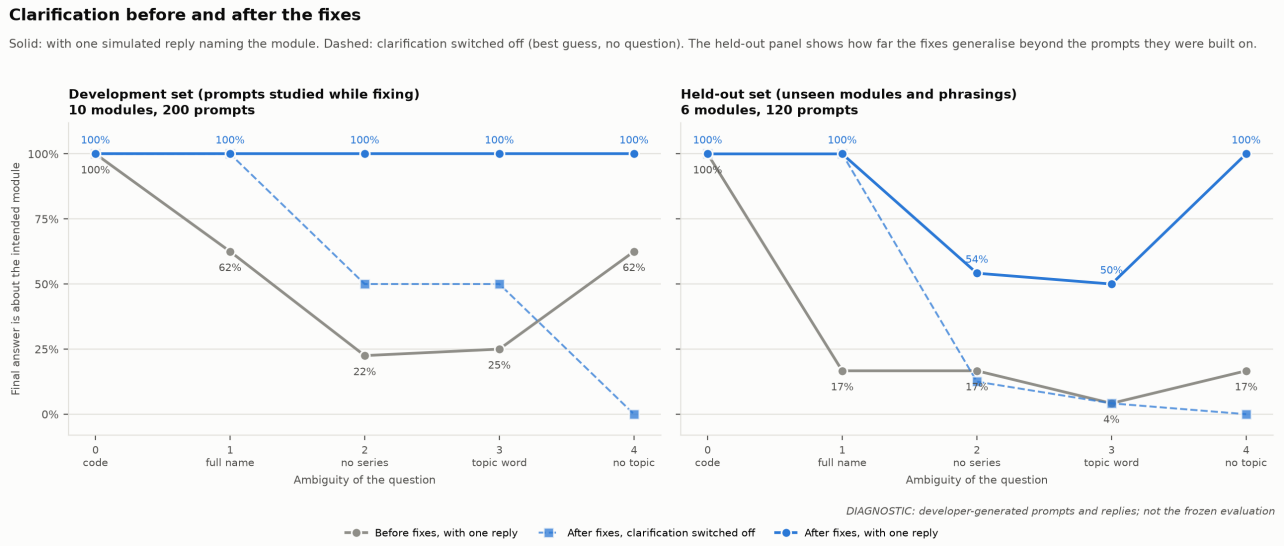


Figure 11. Share of answers about the intended module, before and after the fixes. Left: prompts studied while fixing. Right: unseen module families and phrasings.

The studied set reaching 100% is expected, since the fixes were built against it. The unseen set is the fair measure: large gains, but at the two middle levels only about half succeed. Every remaining failure there is one of two phrasings ("Can you give me info on X", "I need details on X") that the new rule does not recognise, so the rule partly overfitted to the studied phrasings. This is reported rather than patched, because patching it would use up the unseen set.

What clarification itself contributes. The same fixed CurricuPath, with clarification switched off, has to guess the best-scoring module:

Named by	Studied set: off	Studied set: on	Unseen set: off	Unseen set: on
Full name	100%	100%	100%	100%
Name without I/II	50%	100%	12%	54%
Topic word	50%	100%	4%	50%
No topic	0%	100%	0%	100%

With a precise name the two are equal, as they should be. As the question becomes ambiguous, guessing falls away while asking recovers the intended module, including on the unseen set. These are developer-written prompts with a cooperative simulated reply, so they show the mechanism works, not how often real students would answer as simulated.

4.3 Handling typos and unusual wording (interpretation fallback)

When the deterministic interpreter cannot tell what a message means, an interpretation fallback now tries to recover it before the request is refused. It never decides eligibility: it rewrites the message into a standard request (for example "Tell me about ELEN2021A") that the unchanged deterministic pipeline then processes. Prohibited requests and replies to an open question are never reinterpreted.

- **Fuzzy tier** (free, offline): misspelt words close to the catalogue's own vocabulary are corrected.
- **LLM tier:** the model sees the message and the module list and must return one module code from that list, or none; anything else is discarded. Gemini is tried first; Claude Haiku 4.5 is used only when Gemini refuses a call for quota or availability.

It was measured on 34 developer-written cases that were hash-frozen before the fallback code existed, including seven that must not change. The pre-fallback failures were seen before implementation, but the two tiers were specified beforehand and no rules target specific phrases.

Case type	Off	Fuzzy	Fuzzy + LLM
Typos (12)	11	12	12
Unusual phrasing (8)	8	8	8
Typos with shorthand (3)	0	3	3
Descriptions without a name (4)	1	1	3
Must not change (7)	7	7	7
Total (34)	**27**	**31**	**33**

Each tier has a distinct job: fuzzy matching fixed every typo, and only the LLM handled descriptions without a module name. The LLM was called for 2 of 34 cases, so the free Gemini quota goes a long way, and no must-not-change case was reinterpreted at any tier. The 60 eligibility decisions were unchanged. One case still fails: "the course about probability for engineers" is answered as an online-course search, because "course" deliberately does not trigger the fallback.

Caveats: the cases are small and developer-written; the Claude handover is tested only with simulated providers, since no Anthropic key is configured; and with the LLM tier on, student messages are sent to Google (and on handover to Anthropic), so they should not contain personal information.

5. Published benchmarks (context only)

Systems published elsewhere used other institutions, datasets and definitions, so their figures cannot be reproduced or matched here. They are listed beside our measurements for context and are deliberately not drawn on the same axis as our results: a shared axis would imply a parity the data does not support.

Dimension	Measured here (pilot)	Published	Why not comparable
Response time	CurricuPath median 5 ms (deterministic path, no model call); Handbook RAG median 1.6 s (one model call); structured-rules LLM median 1.8 s (one model call)	B6: average query time 2.07 (single institution; Gemini to Cypher over Neo4j)	The published time includes a language-model pipeline. Our CurricuPath figure contains no model call, so only the model-backed baselines are the same kind of measurement.
Decision quality	CurricuPath macro-F1 0.89 (developer labels), 0.65 (briefing labels)	B6: f1 0.958 (300 processed queries listed (420 described); 3 tracks); B1: prerequisite classification 0.99 (synthetic: 101330 interactions, 10000 simulated students; reported as above 99%)	Different task, metric and data. Our answer keys are developer-side, so this measures agreement with our labels, not accuracy.
Intent understanding	85/100 prompts correct (85%); 0/10 on ambiguous module names (bare topic words)	B6: query success 0.93 (curated query collection; single institution)	Different query sets. Ours are unreviewed developer drafts, and the extractor was measured without Gemini.

Published figures with their sources

Benchmark	Metric	Published value	Data behind it
B6 NileAdvisor (Alaa et al. 2025)	average query time	2.07 (seconds)	single institution; Gemini to Cypher over Neo4j
B6 NileAdvisor (Alaa et al. 2025)	f1	0.958 (proportion)	300 processed queries listed (420 described); 3 tracks
B6 NileAdvisor (Alaa et al. 2025)	query success	0.93 (proportion)	curated query collection; single institution
B1 Islam and Hosen (2025)	prerequisite classification	0.99 (proportion)	synthetic: 101330 interactions, 10000 simulated students; reported as above 99%
B5 ARGObot (Tamascelli et al. 2025)	answer correctness	0.878 (proportion)	18 handbook questions
B5 ARGObot (Tamascelli et al. 2025)	faithfulness	0.927 (proportion)	18 handbook questions
B4 MyAdvisor (Kuhail et al. 2023)	student task success	0.92 (proportion)	17 students and 4 advisors
B4 MyAdvisor (Kuhail et al. 2023)	sus students	86.47 (score 0 100)	17 students
B3 Course recommender chatbot (Tan et al. 2022)	usability acceptance	8.2 (score 0 10)	35-student requirements survey
B2 Prerequisite-based recommender (Chanaa and El Faddouli 2024)	ndcg at 10	0.8648 (proportion)	public learning objects; not institutional rules

6. Findings

- **CurricuPath agrees more closely with the developer labels than the handbook-RAG baseline does** (+0.18 macro-F1, interval excludes zero), and points the same way under the briefing labels (+0.14) without reaching a directional result. The direction is stable; the strength depends on the answer key.
- **Against the structured-rules LLM the comparison is inconclusive under both keys** (+0.09 and -0.01). Supplying the rules to a model recovers 48% of its distance from CurricuPath under the developer labels and all under the briefing labels. That supports the value of structured curation, but it does not by itself show that deterministic evaluation caused the gain.
- **CurricuPath and the rules-only system decide identically on 50 of 60 cases.** These cases use clean, fixed wording, so they cannot show the value of language handling. That has to be shown by the ambiguity and clarification experiment. On the other 10 (ELIG031 to ELIG040) the rules-only arm receives the declared completeness fact as a structured field while both CurricuPath arms receive it only as text, and only the rules-only arm decides.
- **The clarification step did not resolve any case in this run.** All 10 cases where clarification was permitted ended unresolved. The scripted reply is a general statement that every module has been listed, and the engine does not read that as declaring the missing corequisite absent. The frozen cases were not edited after seeing system output.
- **Whether planned corequisites count changes the unsafe-error picture.** Under the briefing reading, CurricuPath answers *eligible* on 13 of 53 cases that were not eligible (25%), against 0 under the developer labels. This is the question to put to the academic reviewer.
- **Handbook RAG shows the most unsafe errors** in both keys (28% and 40% of non-eligible cases answered eligible).

7. How these results may be used

Supported by this pilot

- The evaluation pipeline runs end to end: five systems on the same cases, scored with paired bootstrap intervals. This belongs in a methods or pilot section.

- Behavioural findings that do not depend on who wrote the labels: CurricuPath and rules-only decide identically on 50 of 60 cases; on the other 10, CurricuPath (with or without clarification) does not read the declared completeness fact, while rules-only is handed it as a structured field; and Handbook RAG answered *eligible* on non-eligible cases more often than any other system under both answer keys.
- The answer-key sensitivity, which isolates the Rule 1.9 question for the reviewer.
- Latency figures, stated as deterministic-path only for CurricuPath, and entity extraction results, stated as measured on unreviewed prompts.

Not supported by this pilot

- A claim that CurricuPath "achieves" a given macro-F1. That is agreement with developer-side labels, not accuracy.
- Any advantage over the structured-rules LLM. The comparison is inconclusive under both keys.
- Any value from the language layer or from clarification. This case set cannot test them.
- Parity with, or superiority over, published systems. Their figures are context only.

8. Limitations

- Both answer keys are developer-side. The cases and the rules they test were written by the same people, so agreement is partly circular.
- One run per case, so run-to-run repeatability is not measured. The model arms used a lighter model than the pinned one.
- CurricuPath ran without Gemini intent extraction. Its Gemini-assisted behaviour and response time are not measured here.
- 60 cases with about 20 per class give wide intervals. The study can show large failure patterns, not small improvements.
- The intent prompts are unreviewed developer drafts.
- The clarification ablation exists only as a draft run. The reviewer fields it needs are pending, so it cannot yet show whether clarification helps. Grounding of explanations, usability with students and advisors, and recommendation ranking are not evaluated.
- Repeated-failure academic exclusion (for example failing a module twice) cannot be represented by the current student profile, so no case tests it. It is recorded as a known limitation.

9. Next steps

- Record the academic reviewer and grounding judge, with committed hours, in the personnel file.
- Have the reviewer label the 60 blinded cases and draft concealed cases, then run adjudication.
- Repeat the runs with three passes per case on the pinned model, and measure the Gemini-assisted CurricuPath path.
- Ask the reviewer to settle whether a planned corequisite satisfies Rule 1.9, and record the outcome as a rule decision.
- Score grounding and the clarification ablation, then rebuild this report from the official outputs.

Appendix. Reproducing this report

All numbers in this document were read from the analysis files. To regenerate them:

```
py scripts/run_evaluation.py --arm rules_only --allow-draft
py scripts/run_evaluation.py --arm curricupath --allow-draft --runs 1 (with GEMINI_API_KEY blank)
py scripts/run_evaluation.py --arm handbook_rag --allow-draft --runs 1 --pace-seconds 3.5
py scripts/run_intent_evaluation.py
py scripts/run_ambiguity_evaluation.py --allow-draft
py scripts/score_evaluation.py --allow-draft --reference-sets developer briefing
py scripts/render_benchmark_figures.py
```

py scripts/build_evaluation_report.py

Input fingerprints (SHA-256, first 16 characters)

draft_curricupath.jsonl: 76f3d423ba76963a
draft_curricupath_no_clarification.jsonl: 4bba381596176c74
draft_handbook_rag.jsonl: 2138e9d35e3020e3
draft_rules_only.jsonl: e3e9370f9ea0e1b3
draft_structured_rules_llm.jsonl: 5a3c3e4c4ae780bc
eligibility_cases.csv: 955bc130ded299b9
senate_briefing_labels.csv: ab33ec22d9f76467

Answer-key provenance

The briefing labels were extracted from *senate_rules_briefing.docx*, stored beside *senate_briefing_labels.csv* with its hash. The document describes every label as a draft and carries no reviewer name, signature or approval, so it is used for sensitivity analysis only.